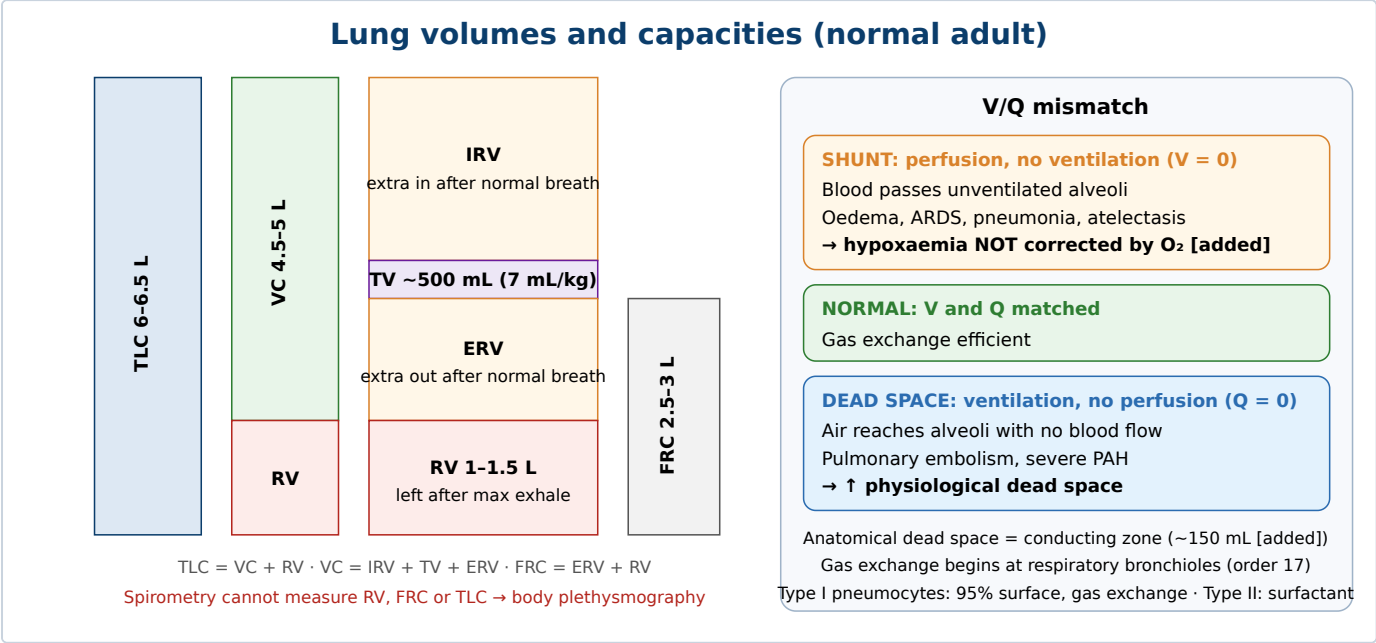


Lung Physiology Primer & Diagnostic Studies (Chest Anatomy & Investigations)

Dr. Essam Kotb, Associate Professor, Internal Medicine | Source: 3-chest_anatomy__invest_-_Updated.pdf (27 slides) · Refs: Harrison 20th ed.; AMBOSS; Cecil 10th ed.; POCUS Atlas

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 **Where this fits:** This is the **foundation lecture** of the respiratory series (lecture 3 in the course order). Every later lecture uses it: **V/Q mismatch** (PE = dead space; pneumonia/oedema = shunt), **lung volumes** (↑ TLC/RV in COPD; ↓ all in restrictive disease), **DLCO** (↓ emphysema and ILD; normal/↑ asthma), **bronchoprovocation** (asthma), and the **PFT algorithm**. If you've already read the COPD, Asthma and Restrictive notes, this is the "why" behind their tests.



Added diagram: lung volumes and V/Q mismatch

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Anatomy: respiratory tract divisions, lobes, mediastinum, pleura	Right 3 lobes / left 2
Knowledge	★ Conducting zone (anatomical dead space) vs respiratory zone	Which generation exchanges gas
Knowledge	★ Type I vs type II pneumocytes	95% vs 5%; gas exchange vs surfactant
Knowledge	★★ V/Q matching: shunt (V = 0) vs dead space (Q = 0)	PE vs pneumonia/ARDS
Knowledge	★★ Lung volumes (TV, IRV, ERV, RV, VC, FRC, TLC) and normal values	Numbers

Domain	Objective	Likely tested as
Knowledge	★★ Obstructive vs restrictive by spirometry, volumes, flow loops	PFT table
Knowledge	★ Plethysmography, DLCO , bronchoprovocation	What each measures
Skills	Correlate exam with pathology; choose imaging/invasive tests; PFT algorithm	Case 1

Flow: Introduction (function, V/Q) → Anatomy (lungs, airway, conducting vs respiratory zone, alveoli) → Lung volumes → Lung function tests (spirometry, plethysmography, DLCO, obstructive vs restrictive, other modalities) → Evaluation of lung disease (imaging, V/Q, bronchoscopy) → Summary algorithm → Case.

2. Main Content (Lecture Order)

2.1 Introduction: Lung Function & V/Q (Slide 4)

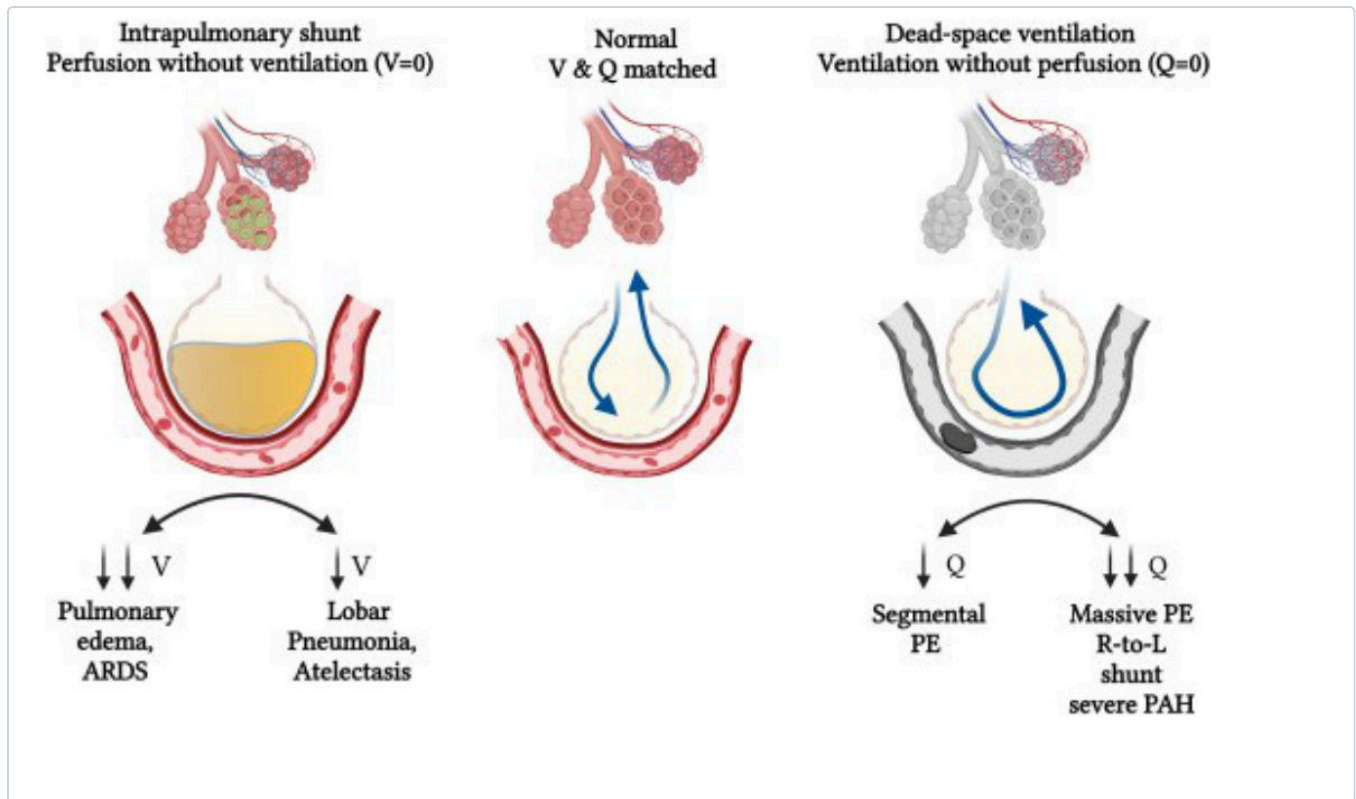
★ **Primary function:** deliver **O₂** to pulmonary capillary blood and **excrete CO₂**, via:

1. **Ventilation (V)**
2. **Perfusion / gas exchange (Q)**
3. ★ **Ventilation-perfusion (V/Q) matching**

★★ **V/Q mismatch (Slide 4 figure):**

	★ Intrapulmonary shunt	Normal	★ Dead-space ventilation
Definition	Perfusion WITHOUT ventilation (V = 0)	V and Q matched	Ventilation WITHOUT perfusion (Q = 0)
Partial (↓)	Lobar pneumonia, atelectasis (↓ V)		Segmental PE (↓ Q)
Severe (↓↓)	Pulmonary oedema, ARDS (↓↓ V)		Massive PE, right-to-left shunt, severe PAH (↓↓ Q)

[added: shunt hypoxaemia responds poorly to supplemental O₂ (the blood never meets the oxygen); dead-space problems respond better. Links: PE pathogenesis in the Pulmonary Vascular lecture.]

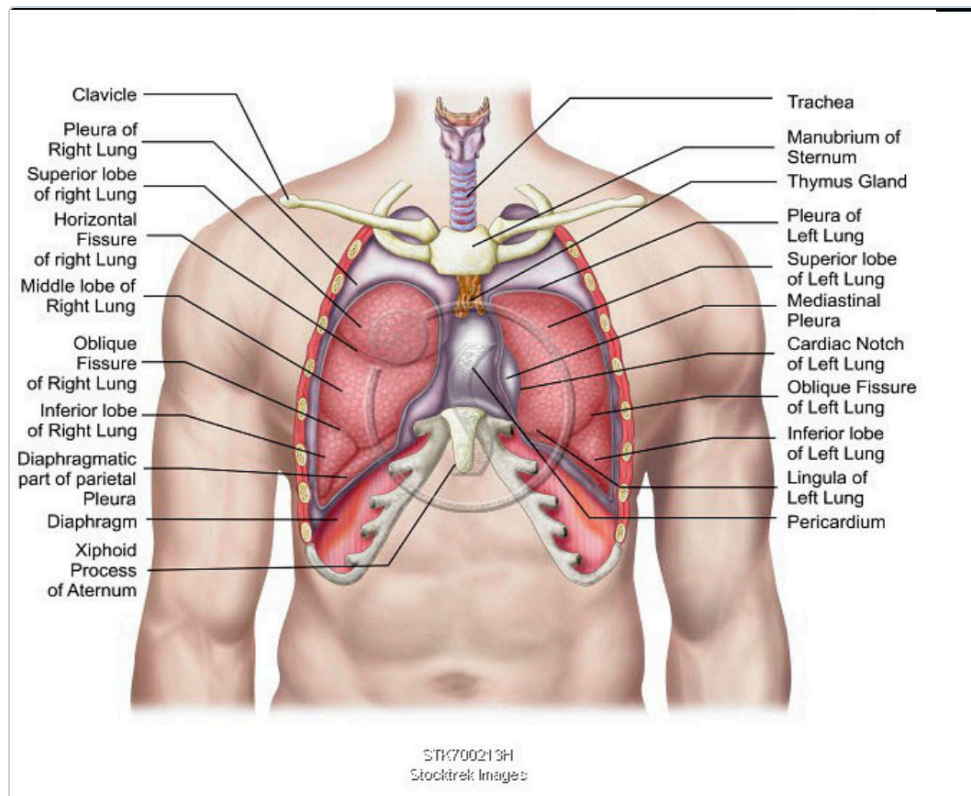


Slide 4: Shunt vs normal vs dead space

2.2 Anatomy (Slides 5-9)

Lungs (Slide 5):

- ★ **Right lung: 3 lobes** (slightly larger); ★ **left lung: 2 lobes**
- Separated by the ★ **mediastinum** (heart, trachea, oesophagus, many lymph nodes)
- Covered by **pleura**; separated from the abdomen by the **diaphragm**

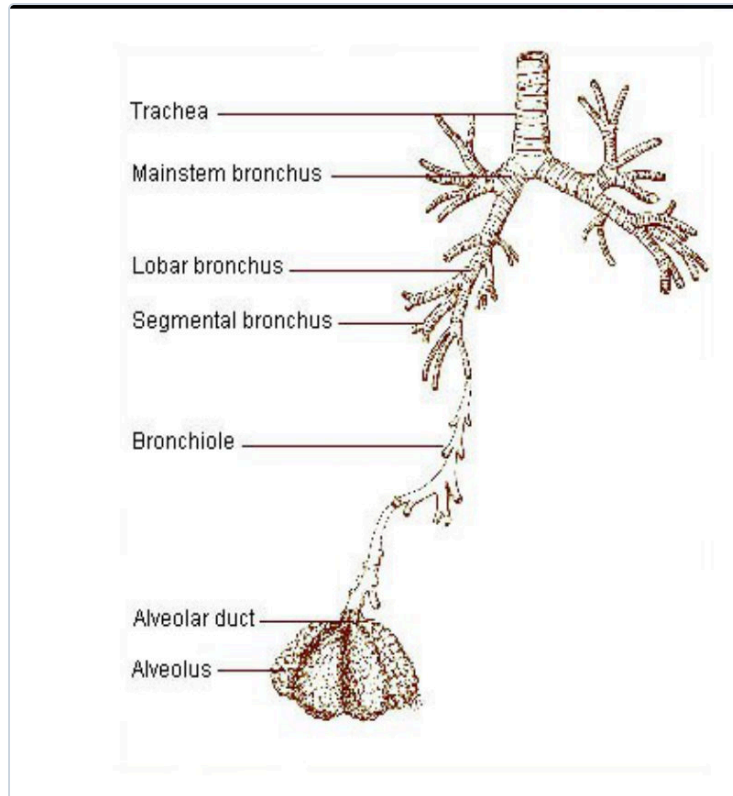


Slide 5: Thoracic anatomy: lungs, lobes, mediastinum

Gas pathway (Slide 6): inhaled air → **trachea** → **bronchi** → **alveoli** (surrounded by capillaries) → O₂ crosses the thin alveolar membrane → **RBCs** carry O₂ to tissues and return **CO₂** → CO₂ released into alveoli → exhaled.

Respiratory tract divisions (Slide 7):

★ Upper	★ Lower
Nose, nasal cavity, pharynx	Larynx, trachea, bronchi, lungs
- Trachea (from the larynx) → two main bronchi → bronchioles → alveoli	
- ★ Alveoli = functional units and site of gas exchange	



Slide 7: Tracheobronchial tree

★★ Airway zones (Slide 8, Weibel table):

Zone	Generations (order no.)	Structures	Key facts
★ Conducting zone = anatomical dead space	0-16	Larynx, trachea (0), bronchi (1-), bronchioles to terminal bronchioles (16)	No gas exchange; cross-section ~2.5 cm ² at the trachea; most resistance in large airways (0.5 vs 0.2 cm H ₂ O·L ⁻¹ ·s in small airways)
★ Respiratory zone	17-23	Respiratory bronchioles (17-19), alveolar ducts (20-22), alveoli (23)	Gas exchange ; cross-section explodes to 5.6 × 10⁷ cm²
[added: small airways contribute little resistance because their total cross-sectional area is huge, so early small-airway disease is "silent" on spirometry.]			

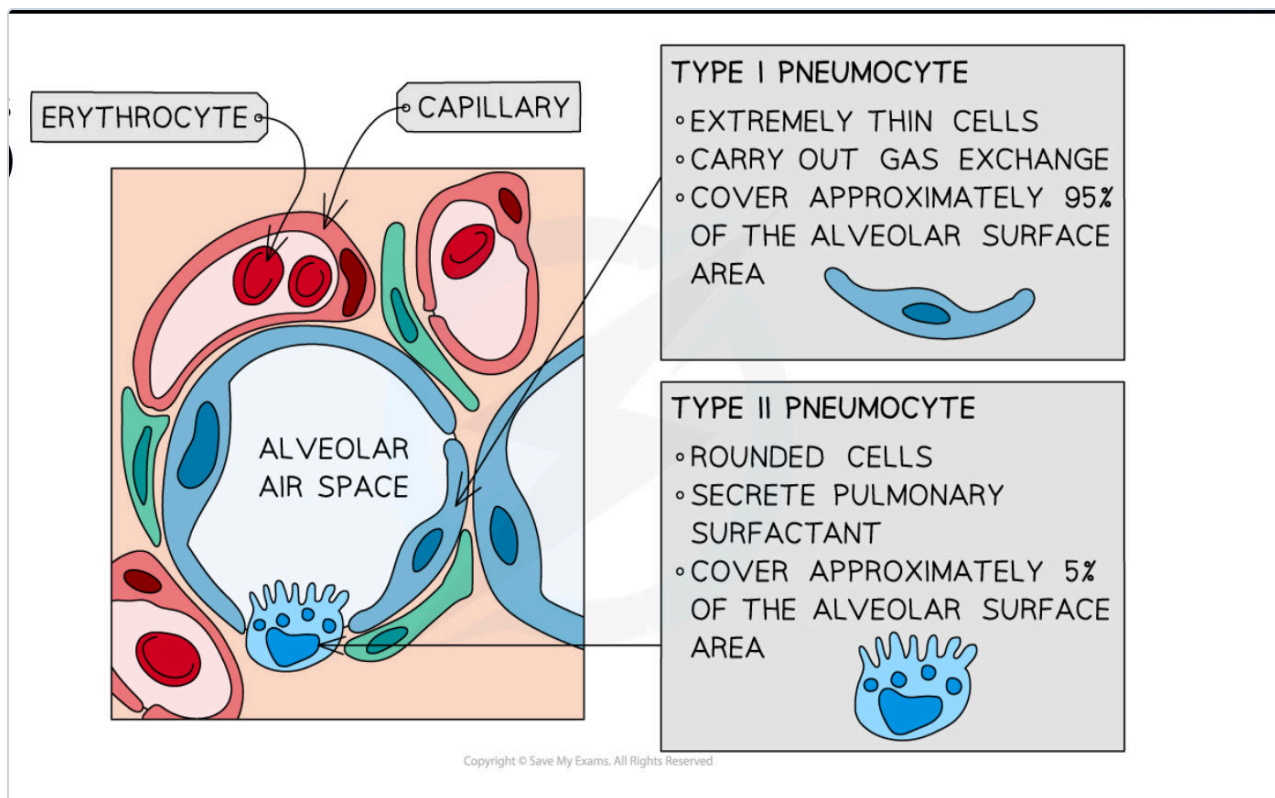
Airway subdivision	Order No.	Cross-sectional area (cm ²)	Resistance (cm H ₂ O • L ⁻¹ • sec)
Larynx	0		0.5
Trachea	0	2.5	0.5
Bronchi	1	2.0	
	2		0.2
Bronchioles		5.0	
	16	1.8 x 10 ²	
Respiratory bronchioles	17		
	19	9.4 x 10 ²	
Alveolar ducts	22	5.8 x 10 ³	
Alveoli	23	5.6 x 10 ⁷	

Fig. 15.1 The subdivisions of the airways and their nomenclature. (Modified from Weibel ER: Morphometry of the human lung, Berlin, 1963, Springer.)

Slide 8: Airway subdivisions, cross-sectional area and resistance

★★★ Alveoli (Slide 9): ★ ~300 million sacs

	★ Type I pneumocyte	★ Type II pneumocyte
Share	95% of alveolar surface	5% of surface
Shape	Extremely thin	Rounded
Function	★ Gas exchange	★ Secrete surfactant [added: and regenerate type I cells; origin of adenocarcinoma (Lung Cancer lecture)]



Slide 9: Alveolus: type I and type II pneumocytes

🔄 **Recap: Anatomy.** Right 3 lobes, left 2. Upper tract to the pharynx; lower tract from the larynx. Conducting zone (generations 0-16) = anatomical dead space; respiratory zone (17-23) exchanges gas. Type I = thin, gas exchange, 95%; type II = surfactant, 5%.

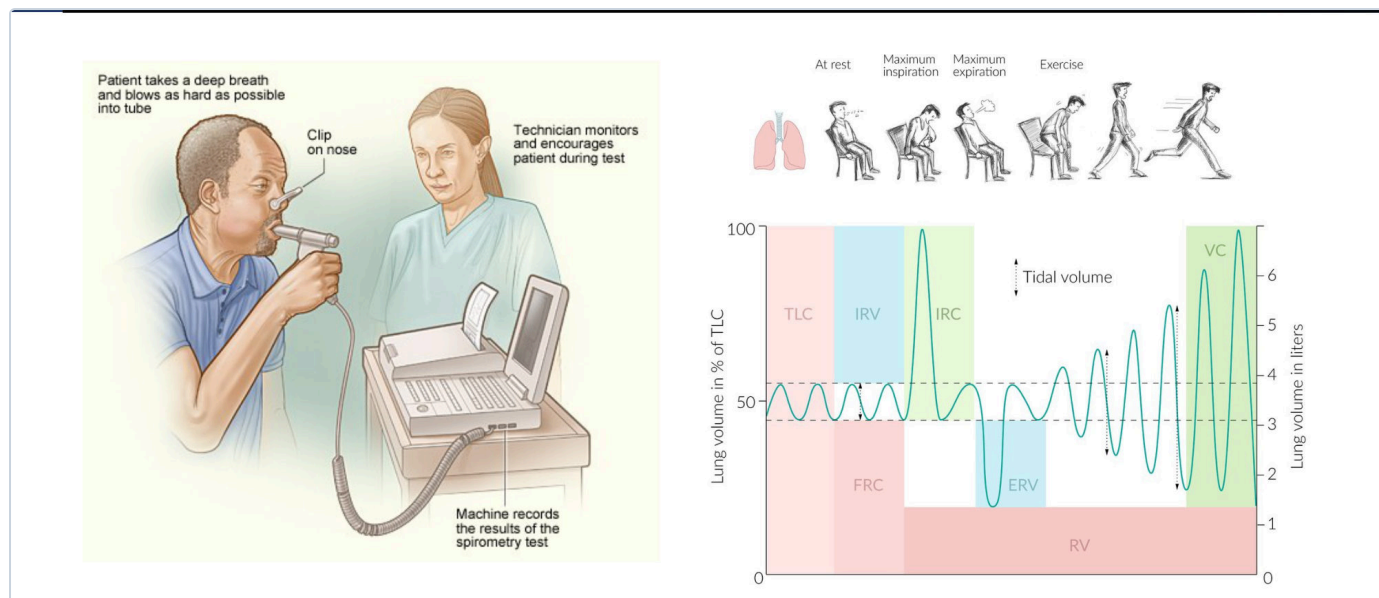
🔗 **Link:** How much air moves in and out is quantified as lung volumes.

👉 **Up next: Lung volumes.**

2.3 Lung Volumes (Slides 10-11)

★★★★ **Normal values:**

Volume	Definition	★ Normal
★ Total lung capacity (TLC)	Air in the lungs after maximal inhalation	6-6.5 L
★ Vital capacity (VC)	Maximal inhalation → maximal exhalation	4.5-5 L
★ Residual volume (RV)	Air left after maximal exhalation	1-1.5 L
★ Tidal volume (TV)	Normal resting breath	~500 mL (7 mL/kg)
★ Functional residual capacity (FRC)	Air left after a normal exhalation	2.5-3 L
<i>IRV and ERV are named in the objectives but not given values on the slides. [added: IRV ≈ 3 L, ERV ≈ 1.2 L; TLC = VC + RV; FRC = ERV + RV.]</i>		



Slide 10: Spirometry: lung volumes trace

2.4 Pulmonary Function Testing (Slides 12-14)

★ **PFTs** measure lung volumes and function; diagnose ventilatory disorders; separate **obstructive vs restrictive**.

Test	What it does
★ Spirometry (most common)	Forced breathing into a device; cheap, simple; measures dynamic and static volumes and flows, ★ EXCEPT RV and TLC
★ Body plethysmography	Measures ★ RV and TLC in a sealed chamber
★ DLCO (single-breath diffusing capacity)	Is the alveolar membrane thickened (fibrosis), destroyed (emphysema), or is the pulmonary vasculature affected (pulmonary hypertension)?

★ **A diagnosis cannot be made from PFTs alone.**

★★★★ **Obstructive vs restrictive (Slide 13):**

	★ Obstructive	★ Restrictive
FEV1	↓ < 80% predicted (or < LLN)	Normal or ↓
★ FEV1/FVC	★ ↓ (FEV1 falls more than FVC)	★ Normal or ↑ (FEV1 falls in proportion to FVC)
VC	↓ < 80% predicted	↓
★ RV (plethysmography)	★ ↑ ≥ 120% predicted (air trapping)	Normal or ↓
★ TLC (plethysmography)	Normal or ↑	★ ↓
★ DLCO	Variable: ↓ emphysema , ↑ or normal asthma , normal chronic bronchitis	Variable: normal extrinsic , ↓ intrinsic

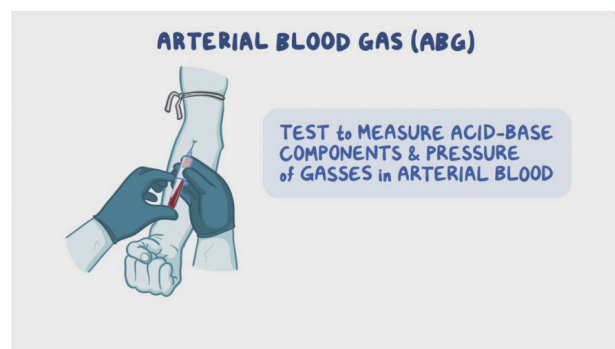
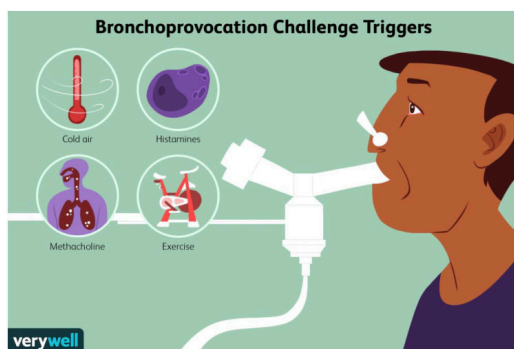
		Obstructive lung disease	Restrictive lung disease
Spirometric findings	FEV ₁	↓ < 80% of the predicted value or the lower limit of normal	Normal or ↓
	FEV ₁ /FVC	↓ (because ↓ in FEV ₁ > ↓ in FVC)	Normal or ↑ (because ↓ in FEV ₁ is proportional to ↓ in FVC)
	VC	↓ < 80% of the predicted value or the Lower limit of normal	↓
Plethysmograph findings	RV	↑ ≥ 120% of the predicted value or the Upper limit of normal	Normal or ↓
	TLC	Normal or ↑	↓
DL _{CO}		Variable (e.g., ↓ in emphysema, ↑ or normal in bronchial asthma, normal in chronic bronchitis)	Variable (e.g., normal with extrinsic causes, ↓ with intrinsic causes)

Slide 13: PFT findings: obstructive vs restrictive

Other modalities (Slide 14):

- ★ **Bronchoprovocation testing** (methacholine/histamine; triggers airway hyperresponsiveness; see Asthma: FEV₁ fall > 20%)
- ★ **Arterial blood gas (ABG)**: acid-base, PaO₂, PaCO₂
- **6-minute walk distance** [added: functional capacity, prognosis in ILD/PAH]

• 6-minute walk distance



Slide 14: Bronchoprovocation triggers and ABG

🔄 **Recap: PFTs.** Spirometry (not RV/TLC) → plethysmography for RV/TLC → DLCO for membrane/vessels. Obstructive: FEV₁/FVC ↓, RV ↑, TLC normal/↑. Restrictive: FEV₁/FVC normal/↑, TLC ↓. DLCO ↓ in emphysema and intrinsic ILD; normal/↑ in asthma; normal in bronchitis and extrinsic restriction.

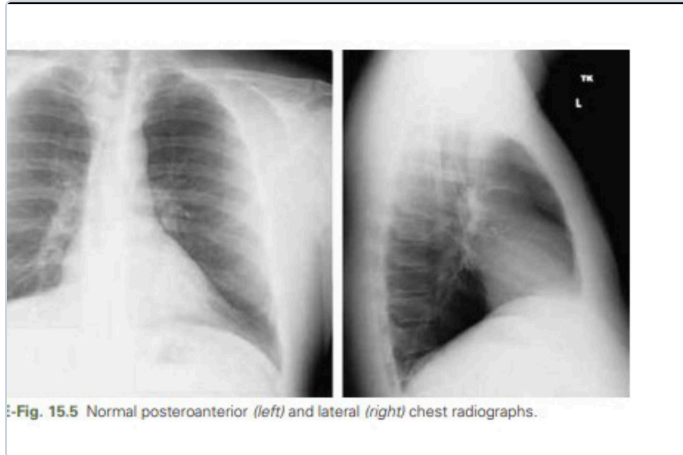
🔗 **Link:** Function tells you the pattern; imaging and invasive tests show the structure.

▶ **Up next: Evaluation of lung disease.**

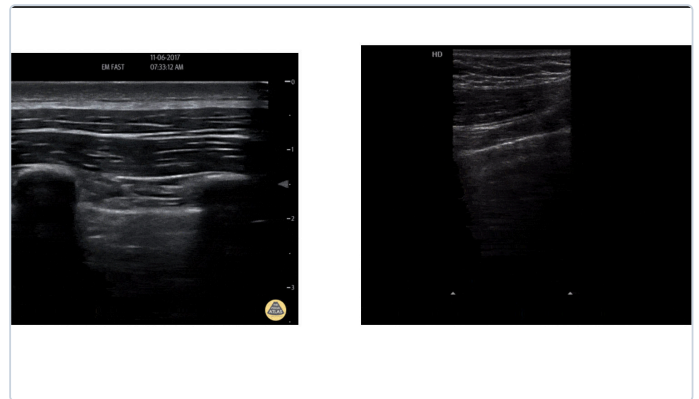
2.5 Evaluating Lung Disease: Imaging & Invasive (Slides 15-22)

★ **Modalities:** CXR, CT, MRI, CTPA · fluoroscopy · ultrasound (POCUS) · pulmonary angiography · V/Q scan · bronchoscopy

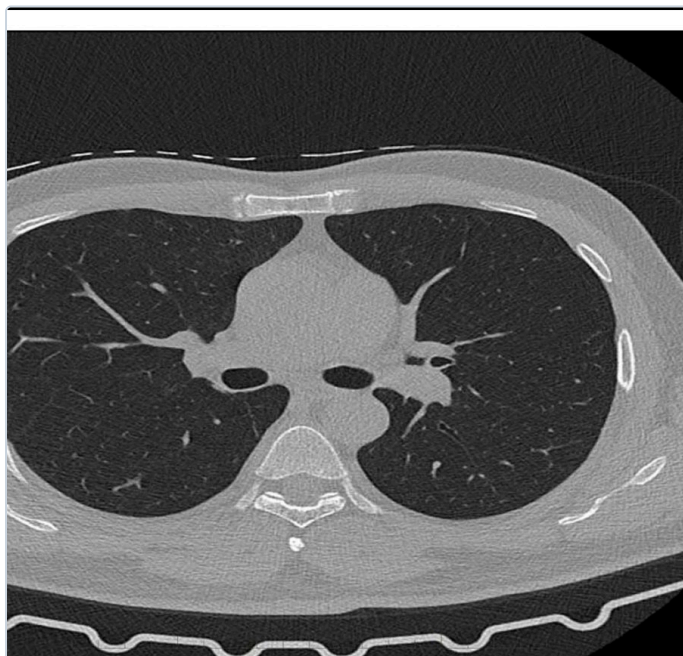
Modality	★ Best use <i>[uses partly added from other lectures]</i>
CXR (PA + lateral)	First-line: consolidation, effusion, pneumothorax, mass, hyperinflation
POCUS	Bedside: effusion, pneumothorax (absent lung sliding <i>[added]</i>), B-lines (oedema)
CT chest / HRCT	Nodules, masses, ILD (honeycombing), bronchiectasis (tram-track/signet ring)
★ CTPA	★ Pulmonary embolism (filling defect)
MRI	Selected: chest wall, Pancoast/superior sulcus, mediastinum <i>[added]</i>
★ V/Q scan	PE when CTPA contraindicated (renal failure, contrast allergy, pregnancy with normal CXR)
★ Bronchoscopy	Visualise airways; biopsy (central tumours, EBUS), BAL, foreign body



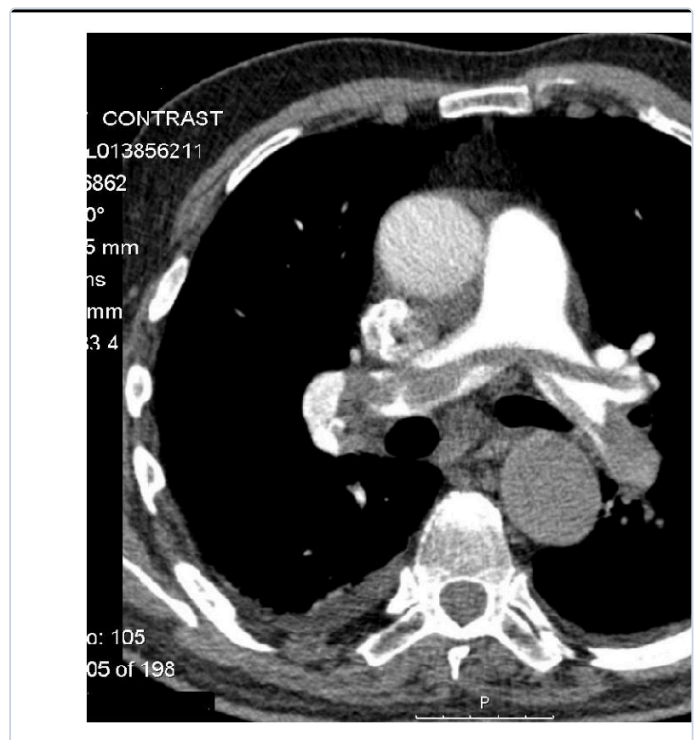
Slide 16: Normal PA and lateral CXR



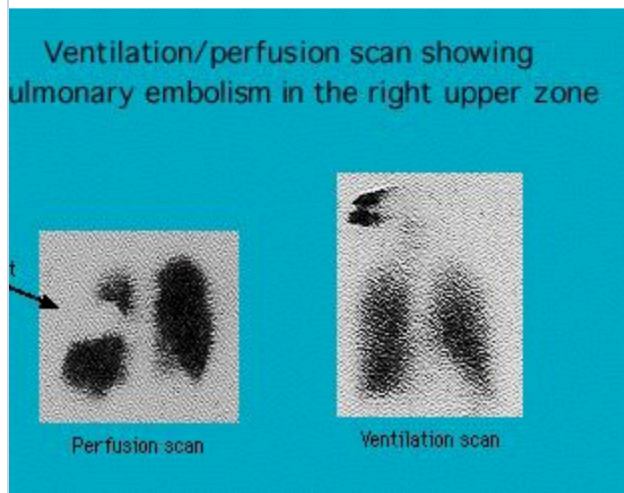
Slide 17: Lung ultrasound (POCUS)



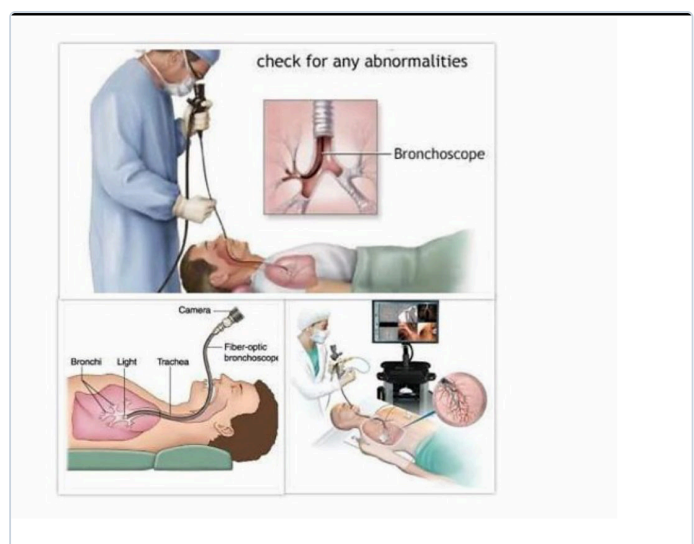
Slide 18: CT chest



Slide 19: CTPA: pulmonary artery filling defect



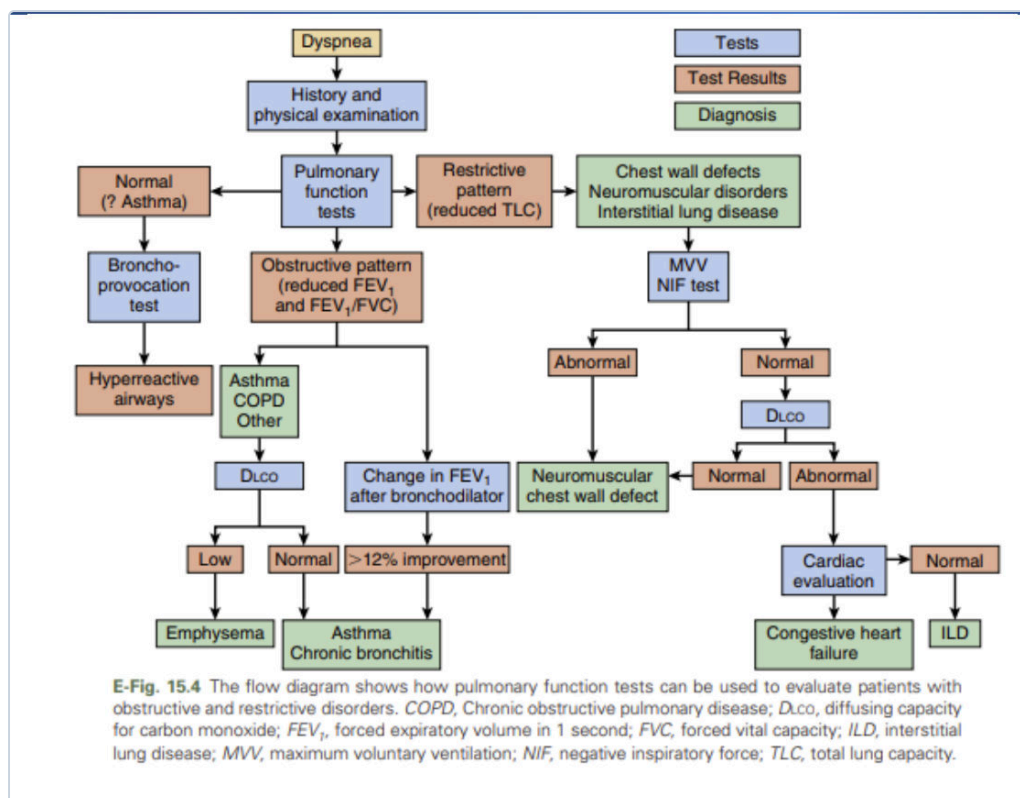
Slide 21: V/Q scan: perfusion defect in the right upper zone with normal ventilation (PE)



Slide 22: Bronchoscopy

2.6 ★★★★★ Summary: The Dyspnoea / PFT Algorithm (Slide 23)

1. **Dyspnoea → history and examination → PFTs**
2. **Normal PFTs (? asthma) → ★ bronchoprovocation test → positive = hyperreactive airways**
3. **★ Obstructive pattern (↓ FEV1 and FEV1/FVC) → asthma, COPD, other:**
 - **DLCO low → ★ emphysema**
 - **DLCO normal → asthma or chronic bronchitis**
 - **FEV1 change after bronchodilator > 12% → ★ asthma (reversible)**
4. **★ Restrictive pattern (↓ TLC) → chest-wall defect, neuromuscular disorder, or ILD:**
 - **MVV / NIF test (muscle strength) → abnormal → neuromuscular / chest-wall defect**
 - **Normal → DLCO:**
 - **Normal → neuromuscular / chest-wall defect**
 - **Abnormal → cardiac evaluation → abnormal = CHF; normal = ★ ILD**



Slide 23: PFT-based dyspnoea algorithm

Recap: Algorithm. Normal PFT → methacholine. Obstruction → DLCO (low = emphysema; normal = asthma/bronchitis) + bronchodilator (> 12% = asthma). Restriction → muscle tests (MVV/NIF) → DLCO → heart → ILD.

3. High-Yield Comparison Tables

Table A: Shunt vs dead space

	Shunt ($V = 0$)	Dead space ($Q = 0$)
Problem	Blood without air	Air without blood
Examples	Pneumonia, atelectasis, oedema, ARDS	PE, severe PAH
O ₂ response <i>[added]</i>	Poor	Better

Table B: Conducting vs respiratory zone (see §2.2)

Table C: Type I vs type II pneumocytes (see §2.2)

Table D: Volumes (see §2.3 and the added diagram)

Table E: Obstructive vs restrictive (see §2.4)

Table F: DLCO

↓ DLCO	Normal DLCO	↑ DLCO [<i>↑ causes partly added</i>]
Emphysema, ILD / pulmonary fibrosis, pulmonary hypertension / PE, anaemia <i>[added]</i>	Chronic bronchitis, asthma (or ↑), extrinsic restriction (chest wall, neuromuscular, obesity)	Asthma , polycythaemia, alveolar haemorrhage

4. Rules, Exceptions, Patterns

- **Right lung 3 lobes, left 2.**
- **Conducting zone (to terminal bronchioles, gen 16) = anatomical dead space; gas exchange starts at respiratory bronchioles (gen 17).**
- **Type I = gas exchange (95%); type II = surfactant (5%).**
- **Shunt = V₀ (pneumonia, oedema, ARDS); dead space = Q₀ (PE).**
- **Spirometry can't measure RV, FRC or TLC; plethysmography can.**
- **FEV1/FVC ↓ = obstruction; normal/↑ with ↓ TLC = restriction.**
- **RV ≥ 120% = air trapping (obstruction).**
- **DLCO separates emphysema (↓) from asthma/bronchitis (normal/↑), and intrinsic (↓) from extrinsic (normal) restriction.**
- **PFTs alone never make the diagnosis.**
- **Normal spirometry + suspected asthma → bronchoprovocation.**

5. Clinical Correlations

- **Hypoxaemic patient with lobar pneumonia not improving on high-flow O₂** → shunt physiology.
- **Sudden dyspnoea, clear lungs, hypoxaemia** → dead space from PE → CTPA (or V/Q).
- **Premature neonate with respiratory distress** → surfactant deficiency (type II pneumocytes) *[added]*.
- **Smoker with FEV1/FVC 0.62, ↑ TLC, ↓ DLCO** → emphysema (Case 1).
- **Obese patient with ↓ TLC but normal DLCO** → extrinsic restriction.
- **Myasthenia/ALS with restriction** → MVV/NIF abnormal (neuromuscular).
- **Asthmatic with normal spirometry between attacks** → methacholine challenge.

6. Rapid Revision (Last-Day)

- Lungs: deliver O₂, remove CO₂ via **ventilation, perfusion, V/Q matching**.

- **Shunt ($V = 0$):** pneumonia, atelectasis, oedema, ARDS. **Dead space ($Q = 0$):** segmental/massive PE, R→L shunt, severe PAH.
- Right lung **3 lobes**, left **2**; mediastinum (heart, trachea, oesophagus, nodes); pleura; diaphragm.
- Upper tract: nose, nasal cavity, pharynx. Lower: larynx, trachea, bronchi, lungs. **Alveoli = functional unit.**
- **Conducting zone (0-16) = anatomical dead space; respiratory zone (17-23).**
- **~300 million alveoli; type I 95% (gas exchange, thin); type II 5% (surfactant, rounded).**
- **TLC 6-6.5 L · VC 4.5-5 L · RV 1-1.5 L · TV ~500 mL (7 mL/kg) · FRC 2.5-3 L.**
- Spirometry: most common; **not RV/TLC. Plethysmography: RV, TLC. DLCO:** thickened (fibrosis), destroyed (emphysema), vascular (PH).
- **Obstructive:** FEV1 < 80%, **FEV1/FVC ↓**, VC ↓, **RV ≥ 120%**, TLC normal/↑, DLCO variable.
- **Restrictive:** FEV1 normal/↓, **FEV1/FVC normal/↑**, VC ↓, RV normal/↓, **TLC ↓**, DLCO normal (extrinsic) / ↓ (intrinsic).
- Other: **bronchoprovocation, ABG, 6-minute walk.**
- Imaging: **CXR, CT, MRI, CTPA**, fluoroscopy, **POCUS**, angiography, **V/Q, bronchoscopy.**
- Algorithm: normal → bronchoprovocation; obstructive → DLCO (low = emphysema) + bronchodilator (> 12% = asthma); restrictive → MVV/NIF → DLCO → cardiac → ILD.

7. Exam-Style Questions

- Q1 (MCQ, lecture Case 1, Slide 24).** 60 ♂, **40 pack-years**, 2 months of progressive exertional dyspnoea and dry cough; **BMI 19** (thin); **prolonged expiration, end-expiratory wheeze**. Post-bronchodilator **FEV1/FVC 62%, FEV1 60%, ★ TLC 125%, ★ DLCO ↓**. Diagnosis? **c. COPD**
Answer: c. [reasoning added; no key on the slide] **Obstruction persisting after bronchodilator** (< 70%) + **hyperinflation (TLC ↑)** + **↓ DLCO** in a thin heavy smoker = **COPD (emphysema-predominant, "pink puffer")**, GOLD 2 (FEV1 60%). **Asthma:** reversible, DLCO normal/↑. **ILD/HP:** restrictive (TLC ↓). **Bronchiectasis:** copious purulent sputum.
- Q2 (MCQ).** Which volume can **not** be measured by spirometry? a) Tidal volume b) Vital capacity **c) Residual volume** d) FEV1 **Answer: c** (also TLC and FRC) → body plethysmography.
- Q3 (MCQ).** A large pulmonary embolism produces which V/Q abnormality? a) Shunt **b) Dead-space ventilation** c) Normal V/Q d) Diffusion limitation only **Answer: b.** Ventilated alveoli with no perfusion ($Q = 0$).
- Q4 (MCQ).** Which cell secretes surfactant? a) Type I pneumocyte **b) Type II pneumocyte** c) Alveolar macrophage d) Club cell **Answer: b.**
- Q5 (Short answer).** A patient has ↓ TLC with normal FEV1/FVC. Muscle strength tests (MVV/NIF) are normal and DLCO is reduced; cardiac evaluation is normal. Diagnosis? **Answer: Interstitial lung disease** (restrictive, intrinsic, normal heart), per the Slide 23 algorithm.

8. Exam Pitfalls & Traps

1. **Shunt vs dead space:** shunt = **no ventilation** (pneumonia, ARDS); dead space = **no perfusion** (PE).
2. **Spirometry measures TLC.** No: **plethysmography** measures RV and TLC.
3. **Obstruction → TLC ↓.** No: **normal or ↑** (air trapping); restriction → TLC ↓.
4. **Restriction → FEV1/FVC ↓.** No: **normal or ↑.**
5. **DLCO ↓ in asthma.** It's **normal or ↑**; ↓ in emphysema.

6. **Type I pneumocytes make surfactant.** Type **II** do; type I do gas exchange.
7. **Gas exchange starts in terminal bronchioles.** It starts in **respiratory bronchioles** (generation 17); terminal bronchioles are still conducting.
8. **PFTs can diagnose alone.** The slide says **no**: combine with clinical findings and imaging.
9. **⚠ Slide title vs file name:** the file is "chest anatomy & investigations"; the slide title is "**Lung Physiology primer & diagnostic studies**". Same lecture.
10. **⚠ Slide 4 labels "Gas exchange (Q)";** Q strictly means **perfusion** (blood flow).